



元智大學
YUAN ZE UNIVERSITY

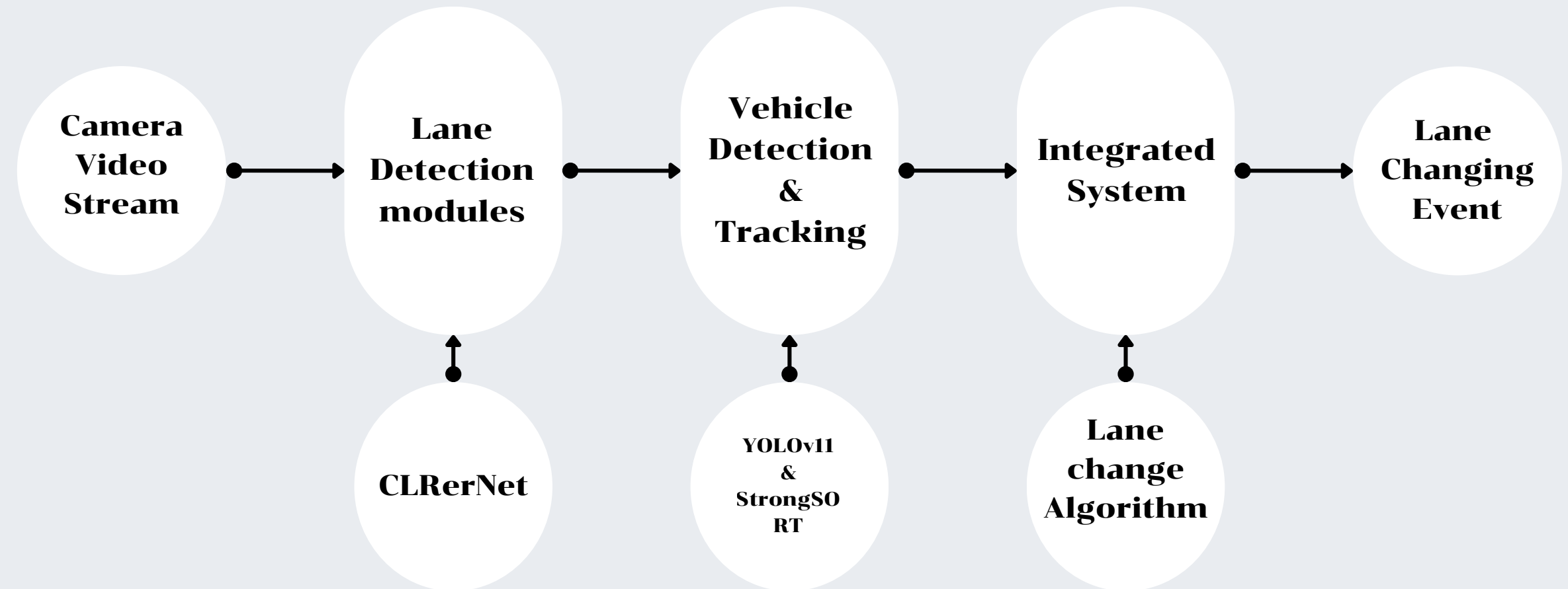
INTERNSHIP RESEARCH REPORT

INVESTIGATE IN IMAGE-BASED VEHICLE
LANE-CHANGE DETECTION

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PROFESSOR:DUAN-YU CHEN

OVERVIEW

- Lane detection
- Vehicle Detection And tracking
- integrated system

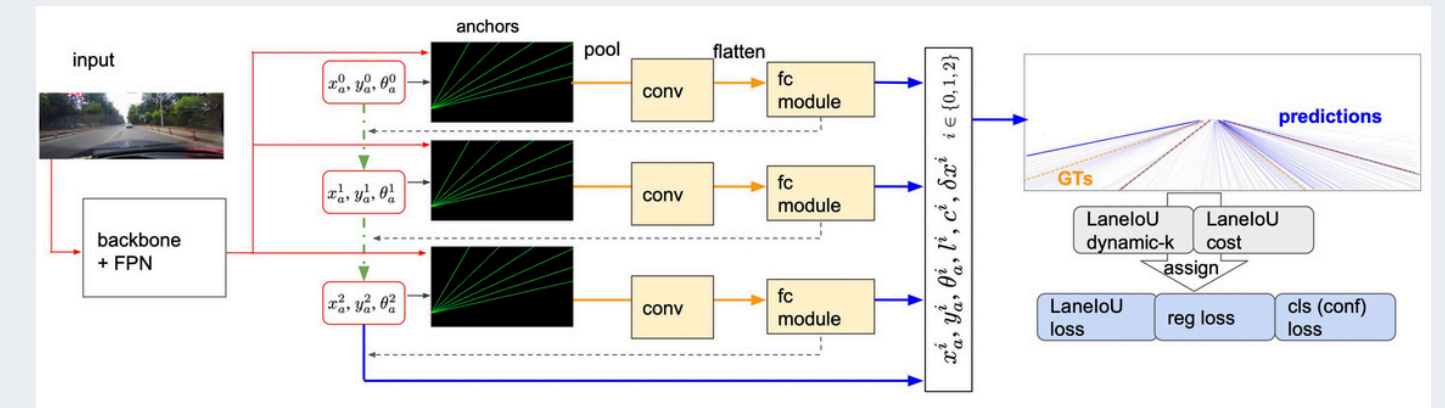


LANE detection

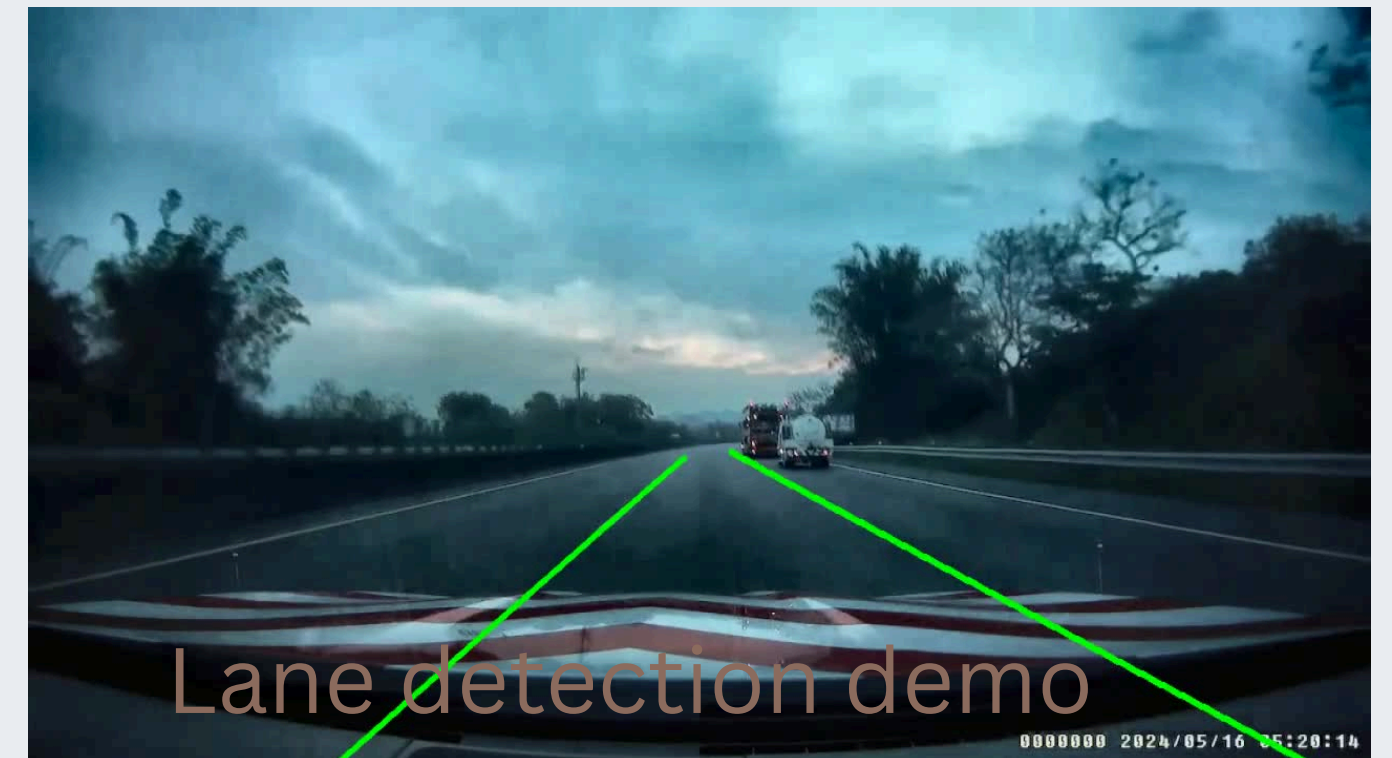
Model : CLRerNET

Method:

- The system uses CLRerNet deep learning model to analyze road images and predict lane line coordinates as ordered point sequences representing lane boundaries.
- Detected lane points are connected and drawn as continuous green polylines on the video frame, providing real-time visual lane boundary detection for intersection



CLRerNetArchitecture



Vehicle detection

Model : YOLOv11-s

Method: The YOLO model analyzes each video frame to detect and classify vehicles (cars, trucks, buses, motorcycles) by predicting bounding box coordinates and confidence scores for each detected object.

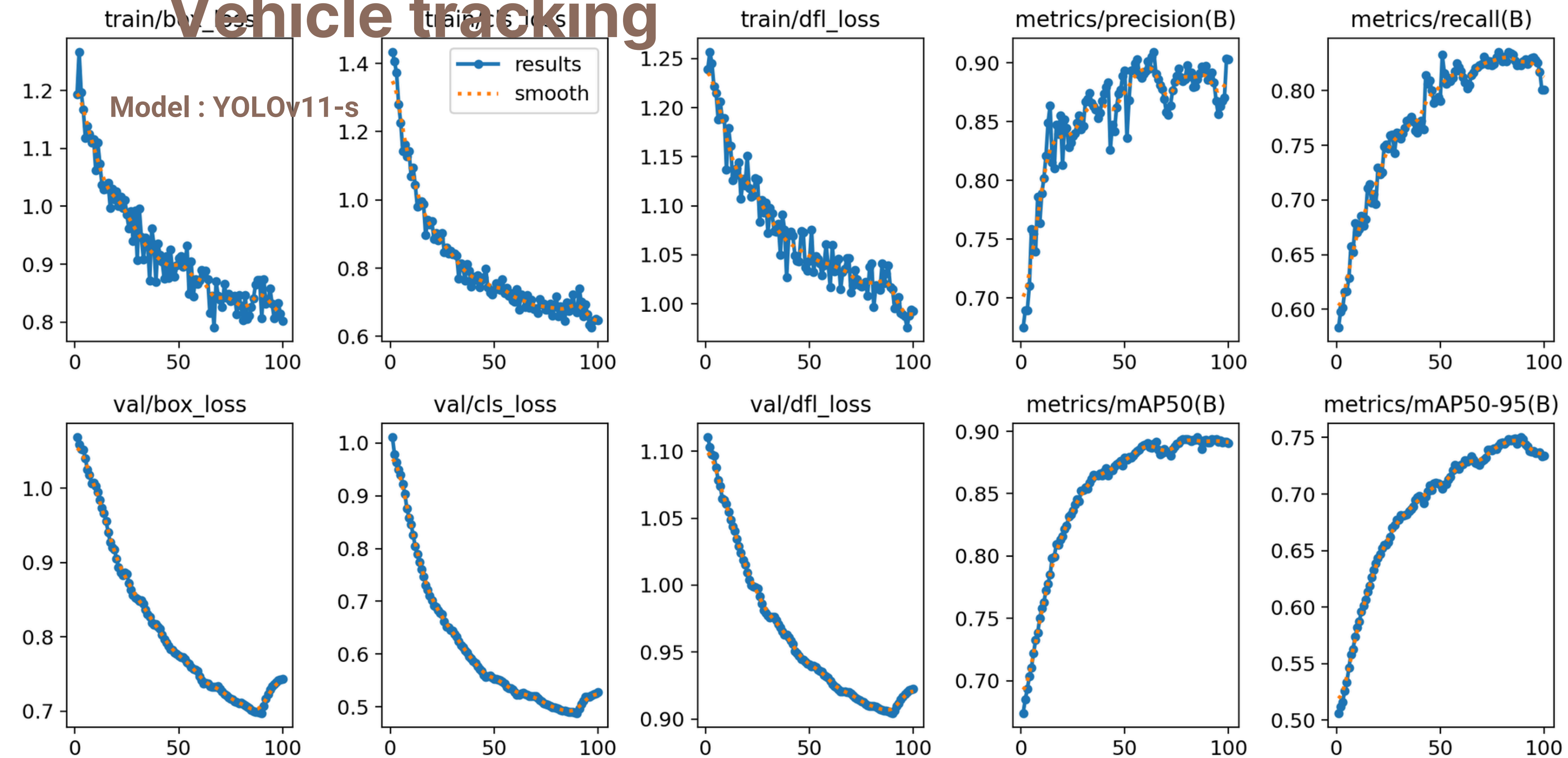
Reason: YOLO has been one of the Best model For Objection task by offer robustness of precision score combining with Strong SORT tracking that offer a continuous detection of the model seamless.



Vehicle detection & tracking demo

Vehicle tracking

Model : YOLOv11-s



Integrated system

Model : YOLOv11-s

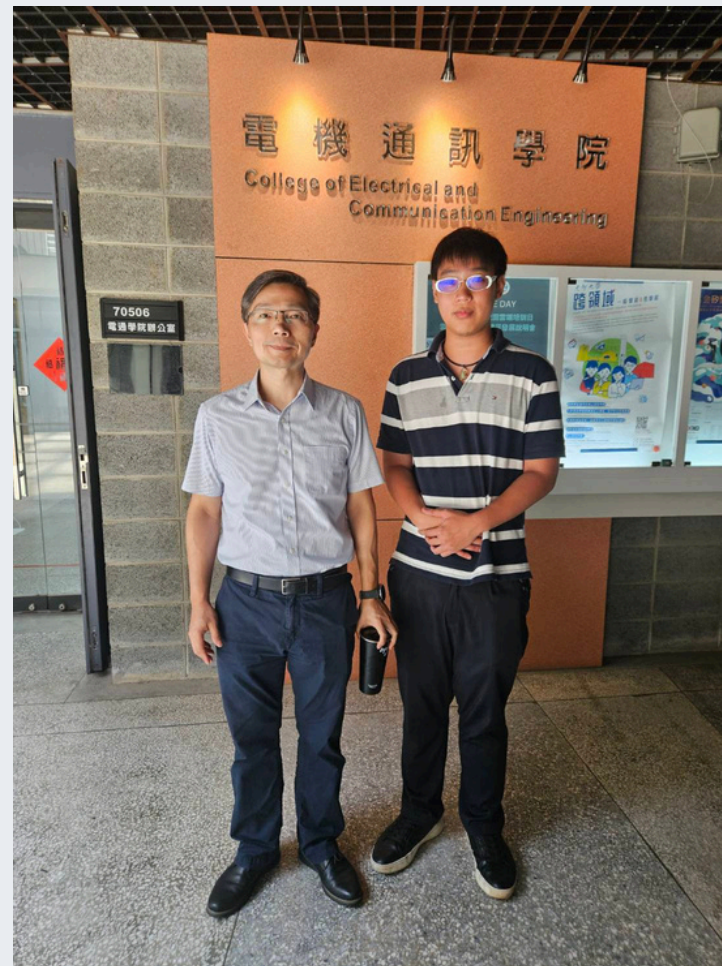
Method:

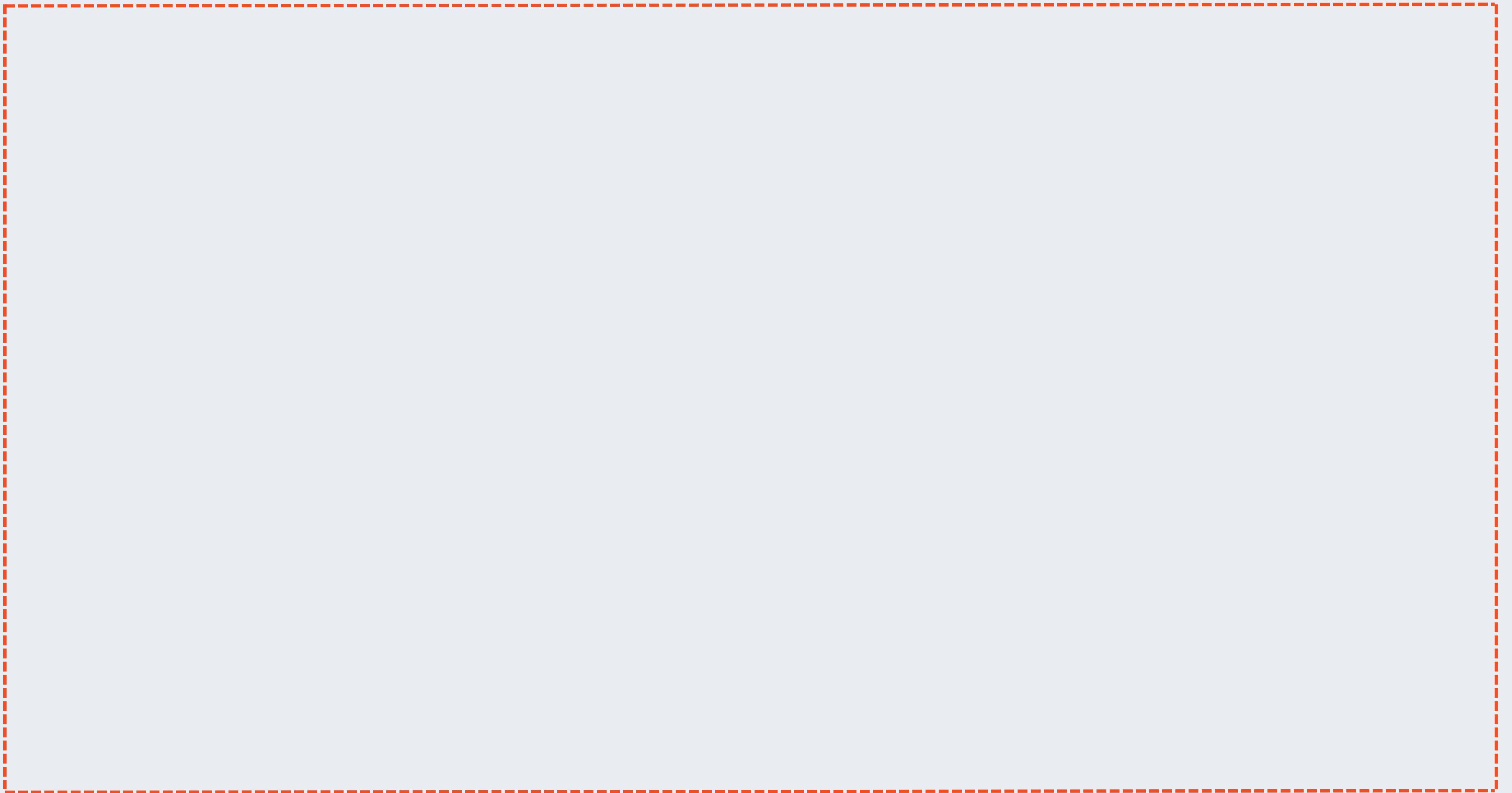
-The system combines YOLO vehicle detection, CLReNet lane detection, and custom tracking to monitor vehicle movement across lane boundaries.

Each vehicle's bottom horizontal line is tracked and when it consistently crosses from one lane to another over multiple frames, a lane change is detected and highlighted.

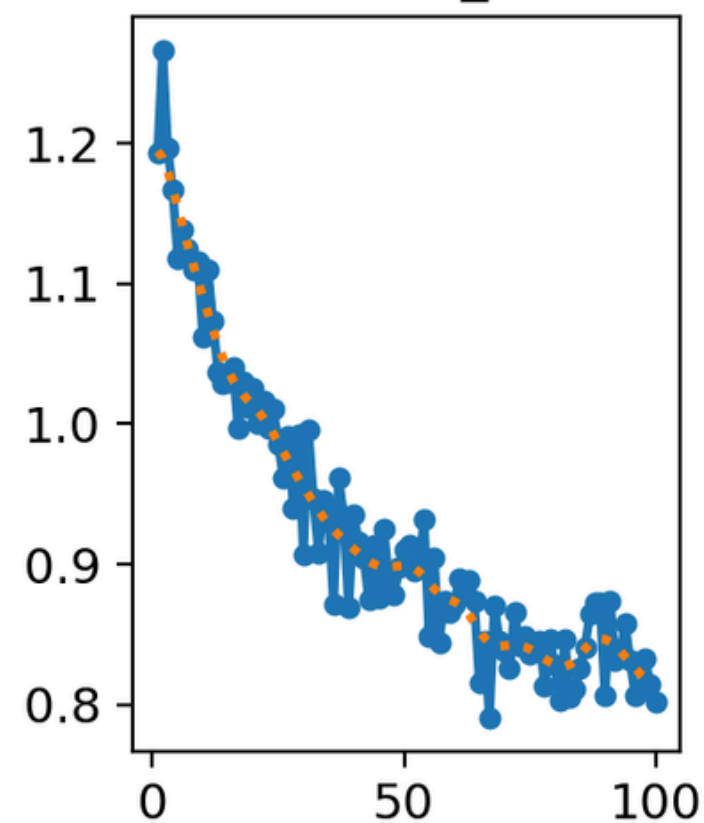


THANK YOU

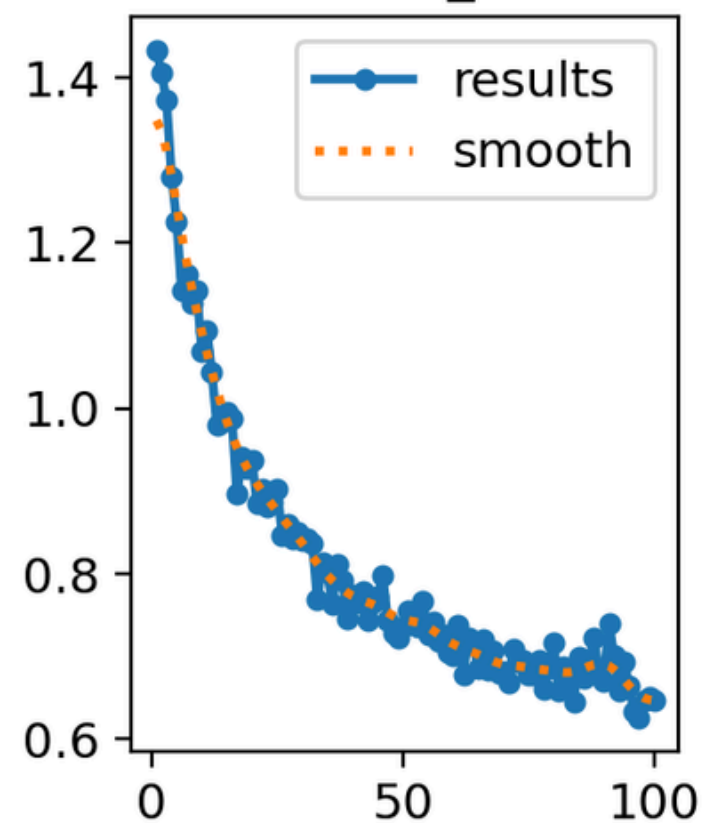




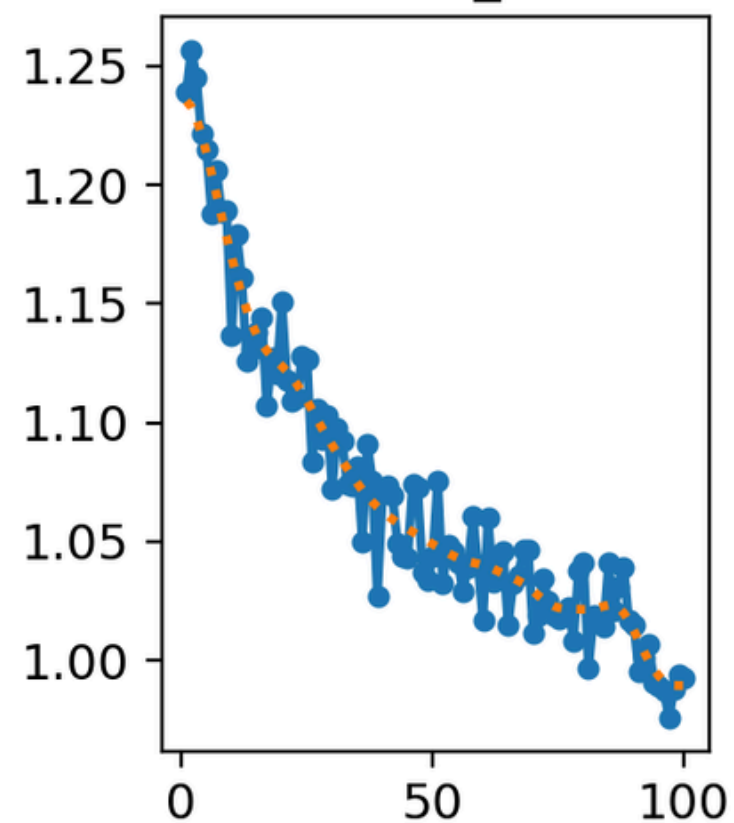
train/box_loss



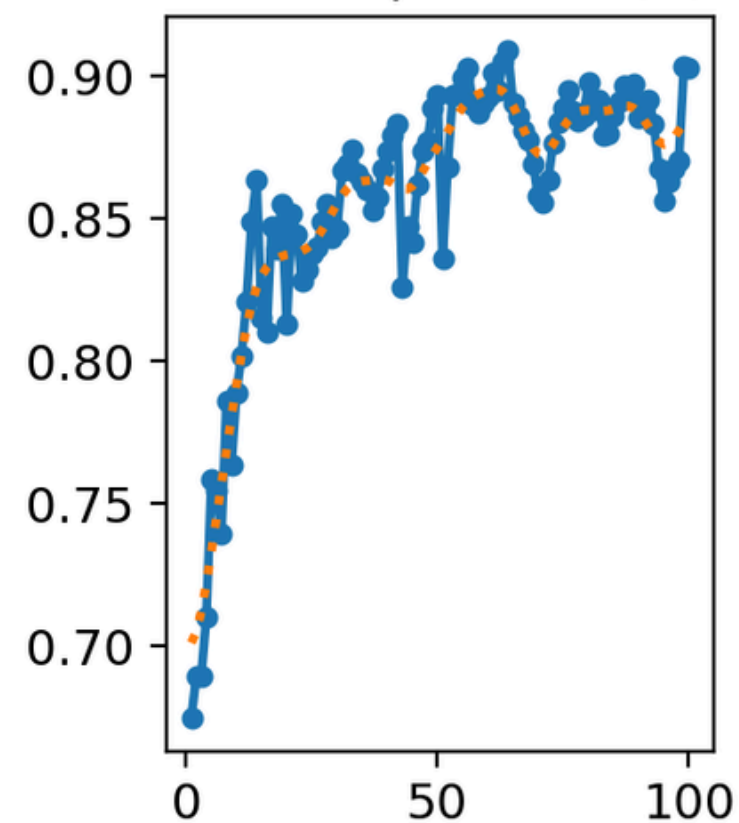
train/cls_loss



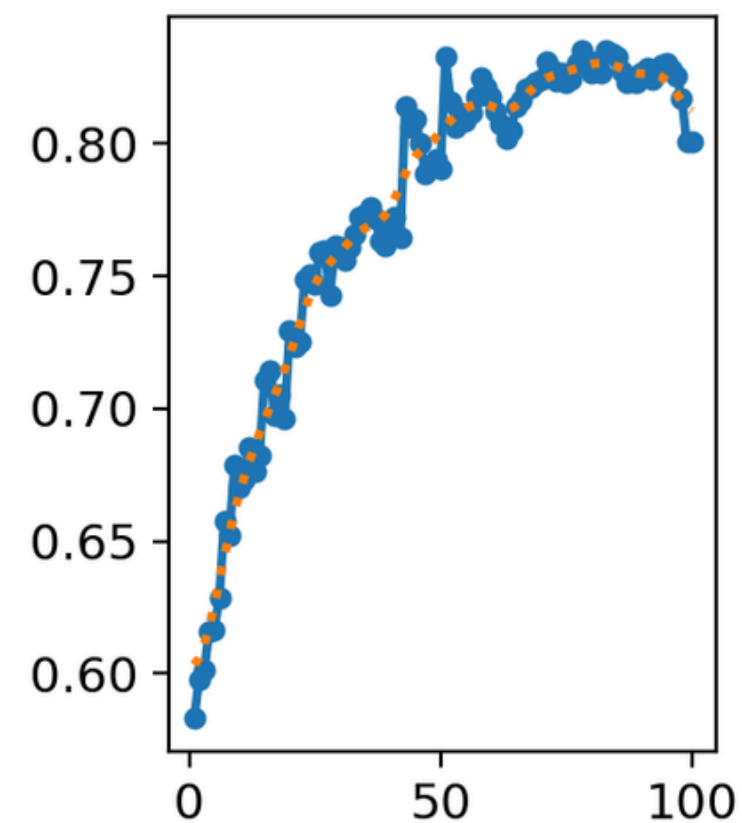
train/dfl_loss



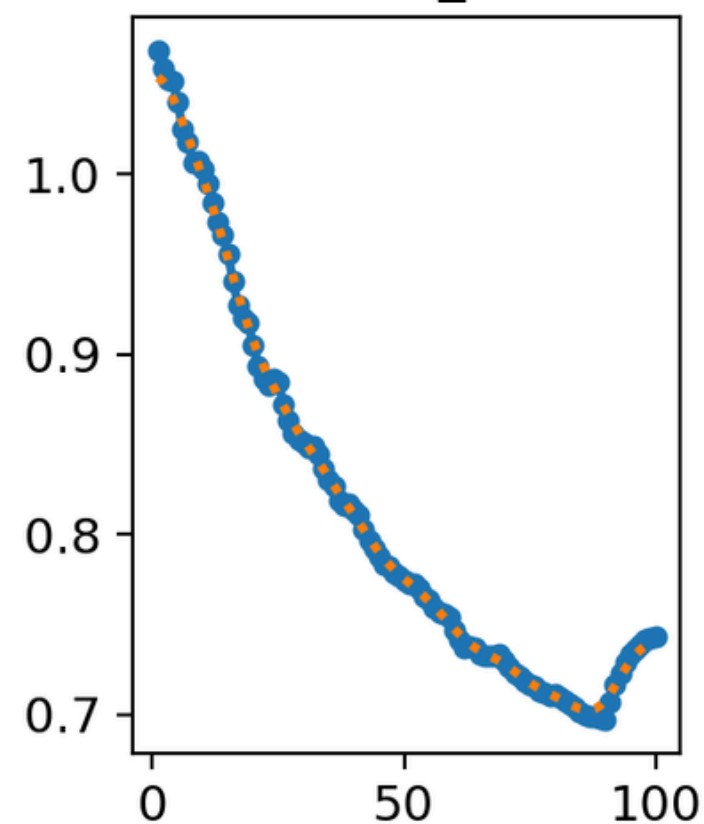
metrics/precision(B)



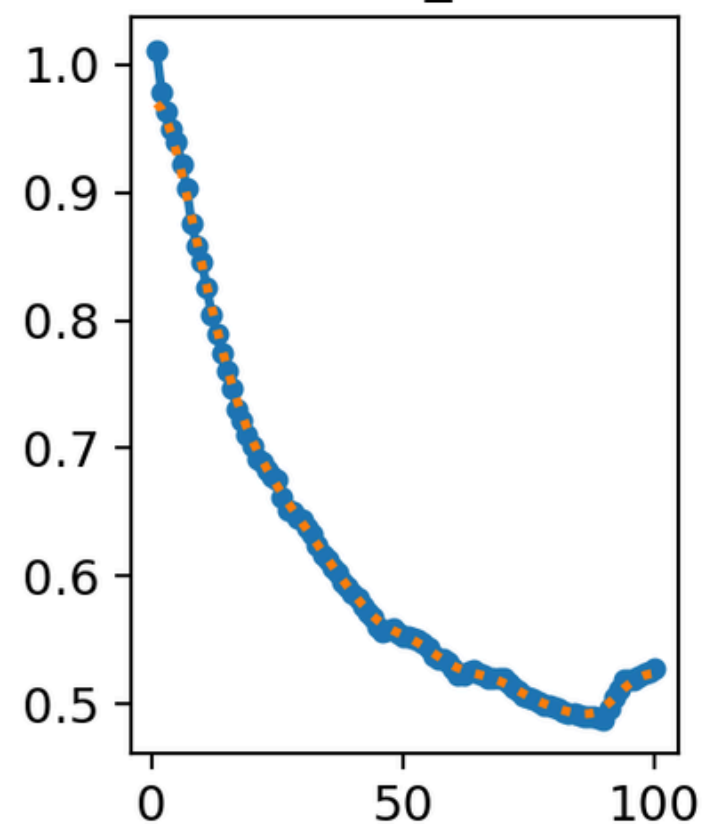
metrics/recall(B)



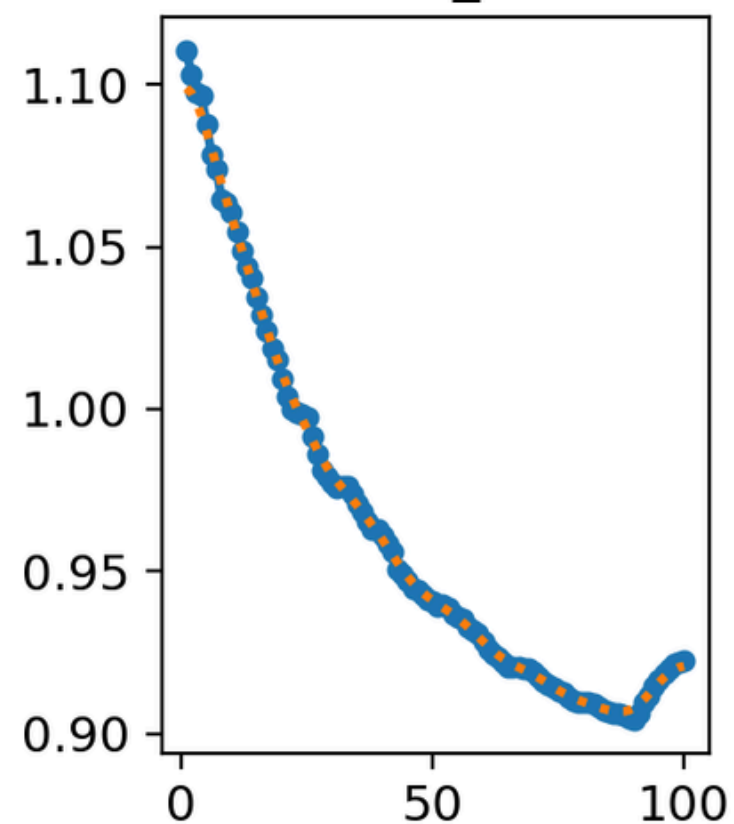
val/box_loss



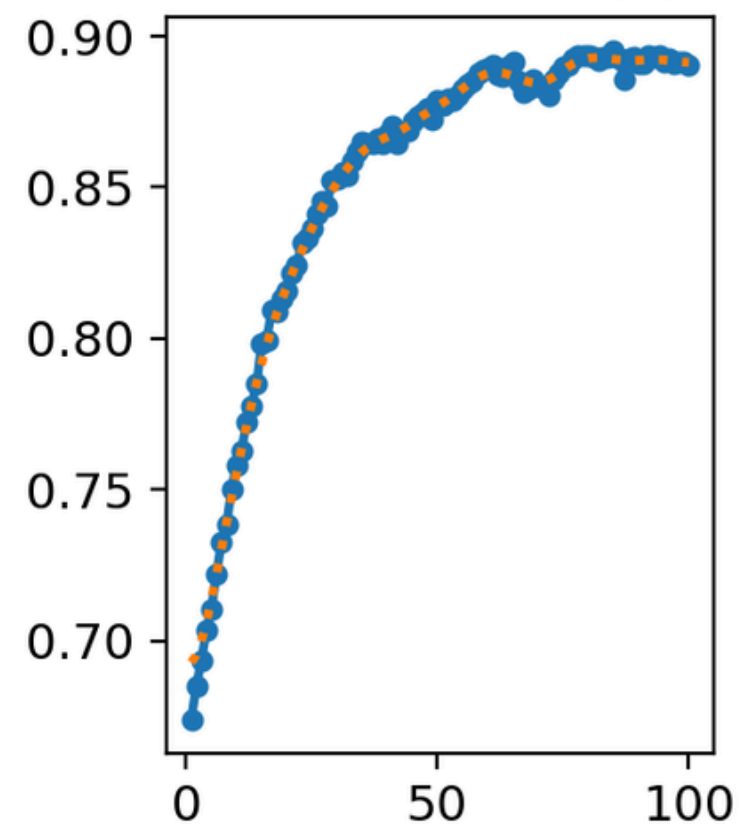
val/cls_loss



val/dfl_loss



metrics/mAP50(B)



metrics/mAP50-95(B)

